

Q1. Can you please provide me a few circuits that I can use to make PCB layouts using some CAD tool?

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A1. I suggest that you make your own schematics and import them in a printed circuit board (PCB) design software to make PCB layouts. All PCB design tools support different schematic entry software. If you take some schematics designed using a schematic entry software that your PCB tool doesn't support, you will not be able to import them to make PCB layouts.

Please follow the steps given below to make a professional-quality PCB:

1. Decide EDA tool chain. First, decide the tool chain to be used for making PCBs. Open Source EDA tools are recommended as their full versions are available free of cost. Two mature Open Source EDA tools are gEDA project and KiCad.

gEDA project. gEDA project is one of the most matured EDA tools that supports complete electronics design cycle, including schematic capture, attribute management, bill-of-materials (BOM) generation, netlist generation, analogue and digital simulation, and printed circuit board (PCB) layout. It

runs on Linux platform.

KiCad. KiCad is an EDA software suite for producing professional-grade schematics and printed circuit boards. Unlike gEDA, it runs on Windows, Linux and Apple OS. KiCad too supports full design cycle, except simulation.

There are several other Open Source EDA tools also but only gEDA project and KiCad have reached the professional-level maturity yet. If you do not want to simulate your circuits before making the PCB layouts, use KiCad. The PCB tool of KiCad is very powerful, allowing even a 3D view of the final PCB.

2. Make schematics. Use schematic entry tool of the tool chain that you have selected for schematic design. 'Eeschema' schematic editor in KiCad can create very good, professional-quality schematics. 'gschem' schematic editor in gEDA project is a powerful tool to register your designs. Symbol libraries of both the tools are equally good and new symbols can be created very quickly.

3. Import schematics to PCB tool. Import schematics created in the previous step to PCB design tool under the same tool chain. Attributes of each component should be carefully entered

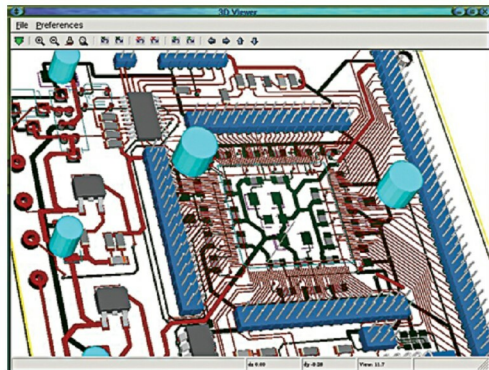


Fig. 1: KiCad 3D Viewer

An ISO 9001:2008 Certified

Crisp System

 Arduino Kit	 Fpga Trainer Kit
 DSP Trainer	 Fingerprint Module
 Force Sensor	 Color Sensor
 GSM Modem	 Voice Recognition Kit
 MP3 Module	 Versa Pad
 RF Transceiver	 Wi-Fi Transceiver
 Accelerometer Sensor Module	 Thermoelectric Power Generator
 Fire Sensor	 Pulse Device
 Humidity Sensor Module	 Pulse Rate Sensor Module
 Voice Recording and Playback Module	 Power line Communication Module
 RFID Reader Module	 Gas Sensor Module

All the above Modules are Ready to Interface with any Micro controller

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Project Guidance also available